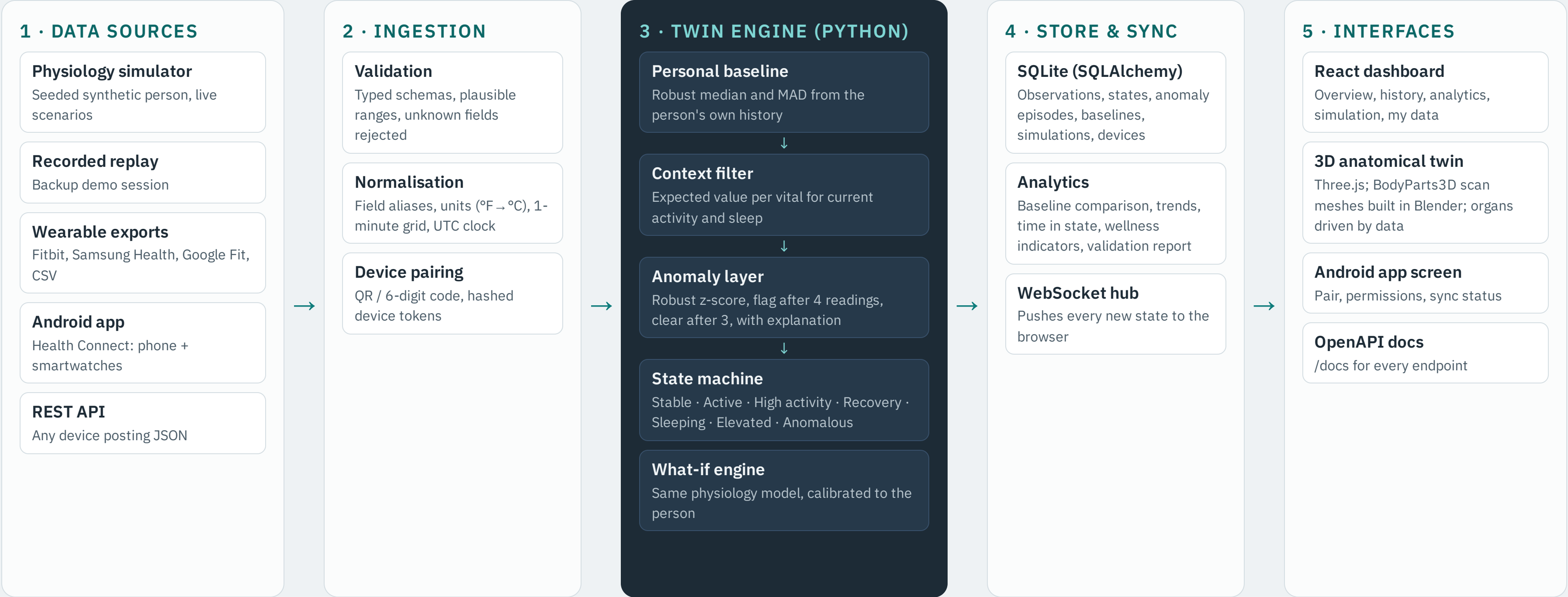


Human Health Digital Twin: system architecture

Data → Twin state → Analytics → Visualisation → Simulation → Feedback. Every source becomes one normalised observation; one engine serves live data, history, uploads, phone sync, what-if and validation.



Feedback loop: users upload data, pair phones, run what-if scenarios and re-learn baselines from the interfaces, and the results flow back through the same engine.

Deployment

One Docker container (API + WebSocket + website): laptop with run.bat, or Render cloud (render.yaml)

Quality

pytest + Vitest; GitHub Actions runs the tests and builds the Android APK on every push

Privacy

Pseudonymous subjects, read-only Health Connect, delete-my-data, not a medical device